Exercises on λ-calculus

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Exercise 1. Give the reduced forms of the following expressions:

$$(\lambda x.x)y \tag{1}$$
$$(\lambda x.yx)z \tag{2}$$
$$(\lambda x.\lambda y.x)(\lambda x.y) \tag{3}$$
$$(\lambda x.xy)(\lambda z.z) \tag{4}$$
$$(\lambda x.\lambda y.xy(xy))(\lambda z.zz) \tag{5}$$

Exercise 2. What does this function test about the number n (using Church encoding)?

$$\lambda n.n(\lambda x.\text{ false})\text{ true}$$

Exercise 3. What is this number (using Church encoding)?

$$\lambda m.\lambda n.nm$$

Exercise 4. Explain why this computes the predecessor of n (using Church encoding):

$$\lambda n.\lambda f.\lambda x.n(\lambda g.\lambda h.h(gf))(\lambda u.x)(\lambda u.u)$$